Emotion Recognition from Photoplethysmography (PPG) Signals

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GithHub Repo

1 Introduction

In this section, we define the scope of this project, the significance of PPG in the context of mobile health, and the specific physiological challenges associated with automated emotion recognition.

1.1 Background

Emotion Recognition is a cornerstone of affective computing, enabling systems to respond to a user's psychological state. While physiological signals like ECG are highly accurate, they often require cumbersome chest straps. PPG, which measures blood volume changes through optical sensors in wrist-worn devices, offers a non-invasive, ubiquitous alternative for long-term health monitoring.

1.2 Problem Statement

The primary challenge in PPG-based emotion recognition is the high susceptibility of the signal to motion artifacts and its lower signal-to-noise ratio compared to ECG. Furthermore, physiological responses to emotions such as amusement and stress can vary significantly across individuals. This project aims to classify emotional states (Baseline/Neutral, Stress, Amusement, and Meditation) using raw wrist-worn PPG data from the WESAD dataset.

1.3 Dataset Overview

For this project, we utilize the **WESAD (Wrist-worn Multimodal Signal Processing for Affect Recognition) dataset**. The study focuses on the Blood Volume Pulse signal sampled at 64 Hz from the wrist sensor. The ground truth labels (sampled at 700 Hz) include 4 emotions: Baseline (Neutral), Stress, Amusement, and Meditation.

2 Data Processing

In this section, we detail the rigorous preprocessing pipeline required to transform raw optical sensor data into a structured format suitable for both traditional and deep learning models.

2.1 Data Extraction

Data is first loaded from subject-specific .pkl files. Since the original WESAD dataset was serialized in Python 2.x, we employ **latin1 encoding** to ensure compatibility with modern Python 3.x environments.

2.2 Signal Preprocessing & Filtering

Raw PPG data from the Empatica E4 is highly susceptible to baseline wander caused by respiration and high-frequency noise from sensor movement. To address this, we implement a 4th-order Butterworth Bandpass filter with a cutoff range of **0.5 Hz to 4.0 Hz**. This specific range is chosen to encompass the physiological limits of human heart rates (30–240 BPM) while filtering out non-cardiac signals.
To ensure that the timing of systolic peaks is not shifted (which would invalidate our HRV calculations), we utilize **sosfiltfilt**. This method applies the filter in both forward and backward directions, resulting in zero phase distortion and a doubling of the effective filter order.

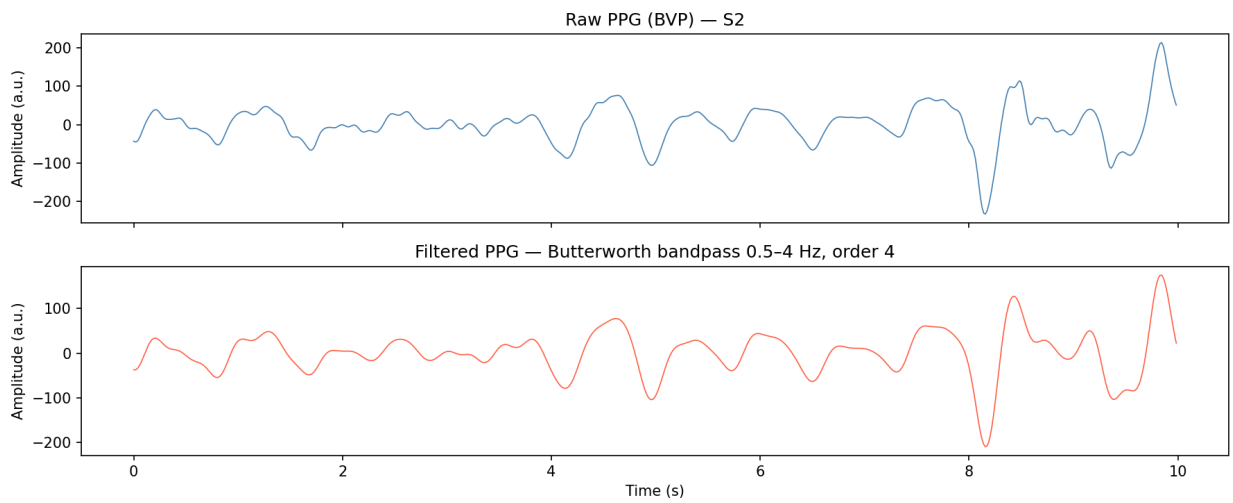


Figure 1. Raw PPG vs Filtered Comparison

Figure 1 illustrates the impact of this preprocessing pipeline. As seen in the comparison, the raw BVP signal (top) exhibits significant high-frequency "jitter" and low-frequency baseline fluctuations. After the application of the Butterworth filter (bottom), the cardiac rhythm becomes more defined, with sharp systolic peaks and a stabilized baseline, facilitating more accurate peak detection and morphological analysis for our downstream models.

2.3 Segmentation and Windowing

Because emotional responses and HRV metrics require time to manifest, we employ a windowing strategy by segmenting the continuous BVP signal into **60-second windows**. A 60-second duration is the minimum standard in physiological literature to derive stable frequency-domain features (such as Low-Frequency and High-Frequency power). In addition, we use a **30-second sliding step** between windows, creating a 50% overlap. This strategy serves as a form of data augmentation, ensuring the models see transitions between states and increasing the total number of training samples.

2.4 Label Assignment

The WESAD ground truth labels are sampled at a much higher rate (700 Hz) than the PPG signal (64 Hz). We first align these by downsampling the labels using nearest-neighbor mapping. To assign a single label to a 60-second window, we use a majority system.
However, windows that occur at the boundary of two emotional tasks can be ambiguous, so a **purity threshold of 0.8** is enforced to maintain data integrity. If the majority label does not constitute at least 80% of the window, the entire segment is discarded, ensuring the model only learns from “clean” emotional states.

2.5 Normalization

PPG amplitude is highly variable between individuals due to differences in traits such as skin tone, subcutaneous fat, and peripheral vascular resistance. To make the data comparable across the 15 subjects, we apply **Z-score normalization** on a per-subject basis. In doing so, we ensure that the models learn the morphology and rhythm of the waves rather than being biased by absolute amplitude differences.

3 Methodology

In this section, we describe the various mathematical and architectural approaches used to classify the processed signals, ranging from traditional statistical features to hybrid deep neural networks.

3.1 Feature-based Approach

We extract a comprehensive set of 18 handcrafted features to represent the physiological state of each window. These include **Time-Domain HRV metrics** that track beat-to-beat variation and **Frequency-Domain HRV metrics** calculated through Welch’s Power Spectral Density. Additionally, we include PPG Morphology features, including skewness, kurtosis, and spectral

power bands, to capture the physical shape of the pulse wave. For classification, we use an **RBF-kernel SVM** and a **Random Forest**.

RBF-kernel Support Vector Machine (SVM). The SVM was selected as a primary classifier due to its proven effectiveness in high-dimensional feature spaces. Because physiological responses to emotion are rarely linearly separable, we utilize the **Radial Basis Function (RBF) kernel**. This kernel uses the "kernel trick" to project our 18-dimensional feature set into a higher-dimensional space, allowing the model to find a complex, non-linear hyperplane that best separates the emotional classes. To optimize the model, we set the regularization parameter C = 10 and gamma = 'scale' *(normalised as 1 / (n_features × Var(X)))* to manage the trade-off between maximizing the decision margin and minimizing the training errors. Additionally, we implemented "balanced" class weights, which assign a higher penalty to misclassifications of minority classes. This ensures the SVM does not simply default to predicting the "Baseline" class (which is more prevalent in the WESAD dataset).

Random Forest Classifier. The Random Forest was employed as a robust ensemble learning method, which works by constructing a multitude of decision trees during training. For this project, we utilized a forest of 300 trees *(max_depth=None, max_features=sqrt(n_features), class_weight='balanced')* to ensure stable and reliable predictions across the 15 subjects. The Random Forest is particularly well-suited for PPG data because it is inherently resistant to outliers and noisy data points. Unlike the SVM, it is non-parametric and does not require feature scaling. A critical benefit of the Random Forest in our analysis was its "Feature Importance" output, which provided clinical interpretability by revealing which physiological markers were most influential in identifying emotional shifts.

3.2 End-to-End Approach

Unlike traditional models, end-to-end neural networks are better for handling the noisy nature of wrist-worn PPG data. These neural networks ingest raw, filtered BVP waveforms and autonomously learn a hierarchy of features. This eliminates the need for manual feature engineering and allows the models to discover physiological patterns that may be invisible to standard statistical calculations. For this project, we implemented a **1D Convolutional Neural Network** focused on spatial morphology and a **CNN-LSTM Hybrid** designed to capture complex temporal dependencies.

1D Convolutional Neural Network. Our 1D CNN architecture is designed to act as an automated feature extractor. By using three successive convolutional blocks with varying kernel sizes, the model performs "feature learning" directly on the 3,840 samples in each 60-second window. The initial layers with larger kernels focus on the broad, low-frequency components of

the pulse wave, while deeper layers with smaller kernels identify high-frequency morphological details, such as the dicrotic notch. This end-to-end structure is particularly resilient to the noise present in the WESAD dataset, as it can learn to prioritize the rhythmic structure of the signal even when traditional peak-finding algorithms might fail.

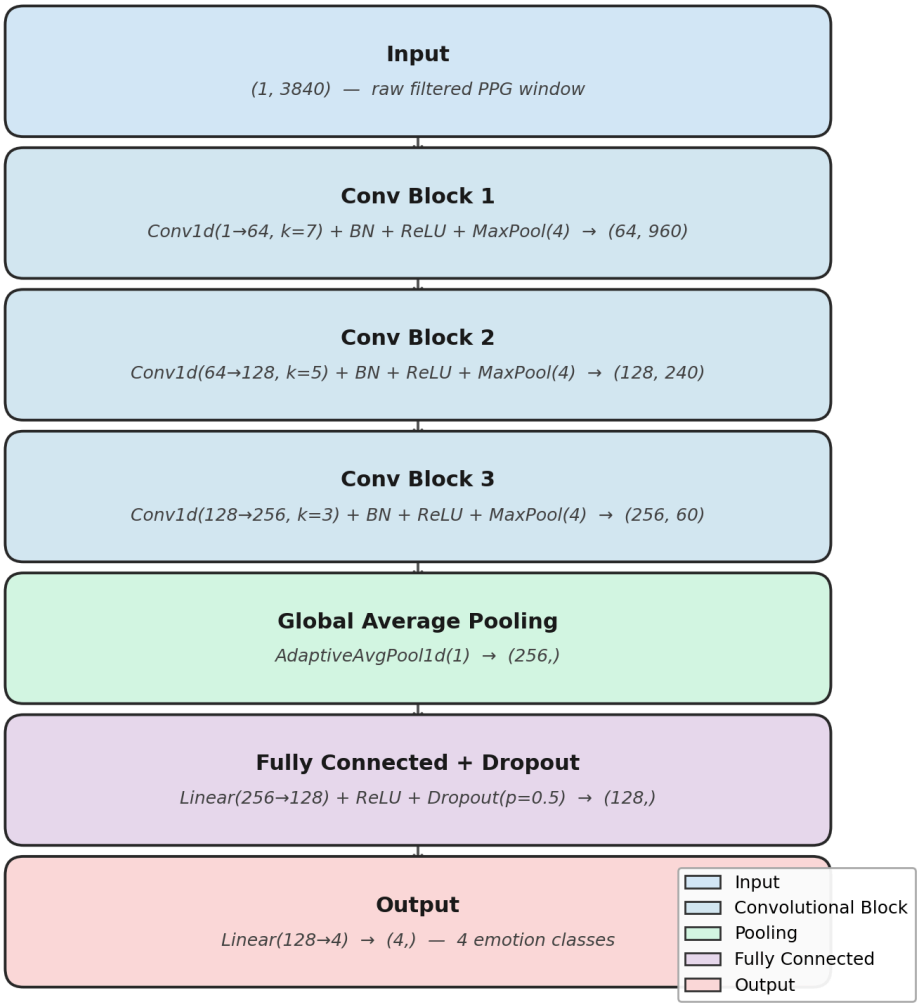


Diagram 1: Architecture of the 1D CNN model. The network processes a raw 60-second PPG window (3,840 samples) through three convolutional blocks with decreasing kernel sizes (7→5→3) and increasing filter depths (64→128→256), followed by global average pooling and a fully connected classifier with dropout regularization.

CNN-LSTM Hybrid Architecture. To capture the evolving nature of emotional states, we implemented a CNN-LSTM hybrid model that treats the PPG signal as a time-series of cardiac events. In this architecture, a **"Time-Distributed" CNN** first processes the window in one-second chunks, distilling raw voltage changes into meaningful cardiac feature vectors. These vectors are then fed into a **Long Short-Term Memory (LSTM) network**, which is uniquely capable of remembering information over long durations. This allows the model to "track" how

physiological markers—such as the gradual increase in heart rate or the loss of rhythmic regularity—unfold over the 60-second window, providing a sophisticated temporal analysis that mirrors the way human emotions actually develop and persist over time.

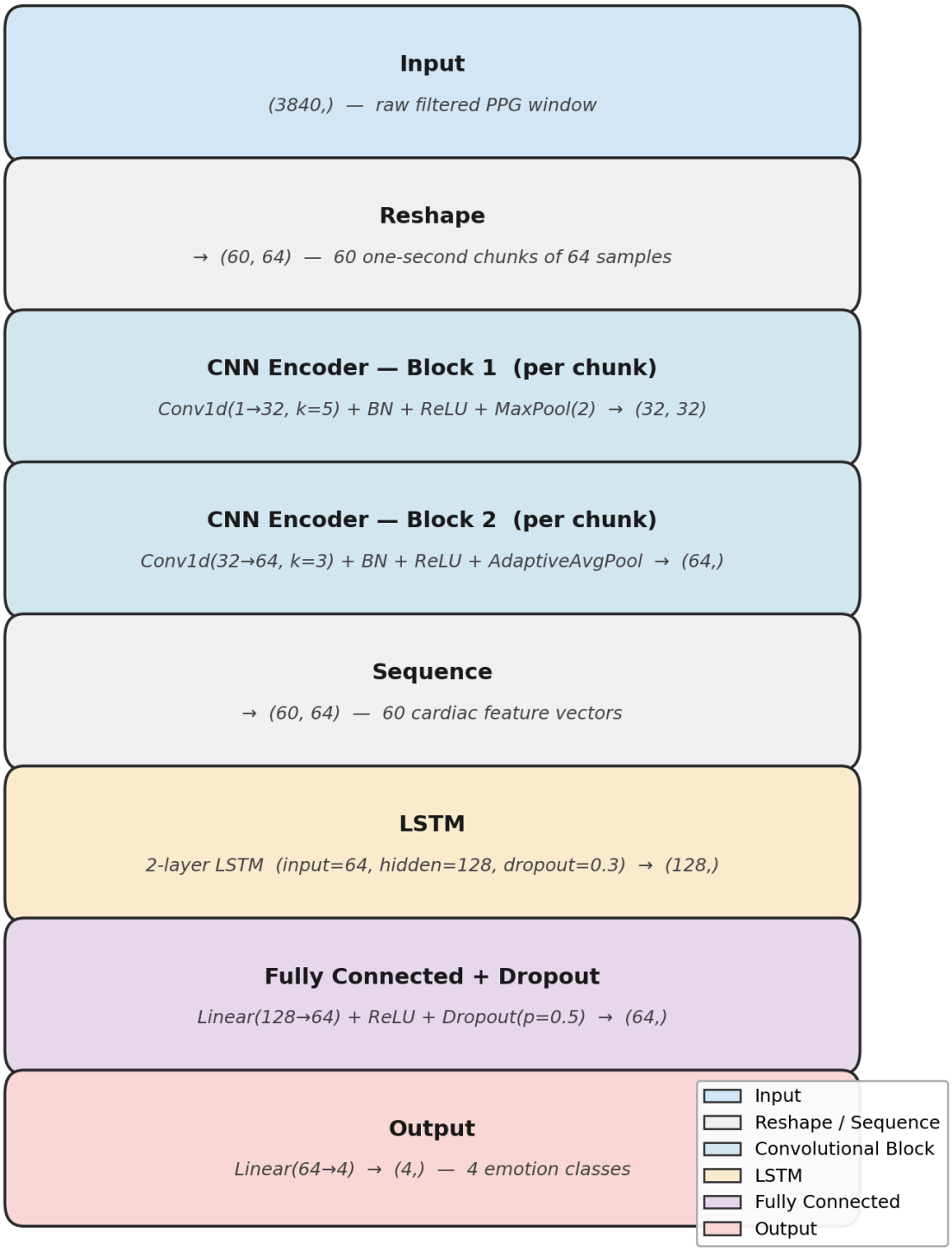


Diagram 2: Architecture of the CNN-LSTM hybrid model. The network divides each 60-second PPG window into 60 one-second segments. A small CNN processes each segment individually to

extract a compact set of physiological features. These features are then passed to a 2-layer LSTM, which learns how these features change over the full 60 seconds to predict the emotional state.

3.3 Train/Test Split Strategy

To validate our models, we use subject-independent protocols, specifically focusing on the transition from a fixed-subject split to the more rigorous LOSO cross-validation.

Fixed Subject-Independent Split. For our initial experiments and deep learning models, we held out three subjects (S2, S3, and S4) to be our final test set, while the remaining 12 subjects were used for training. This represents a roughly 80/20 split of the participants. This strategy was chosen for the deep learning models because training neural networks is computationally expensive. In comparison, a fixed split allows for faster iteration while still maintaining the "unseen user" requirement.

Leave-One-Subject-Out (LOSO) Cross-Validation. To provide a more statistically robust evaluation of our feature-based models, we implemented LOSO cross-validation. In this protocol, the model is trained 15 separate times (once for each subject in the dataset). In each fold, 14 subjects are used for training, and the last subject is held out as the sole test case. This ensures that every single participant in the dataset eventually serves as the test set. By interpolating between existing minority samples rather than simply duplicating them, SMOTE prevents the classifier from developing a bias toward the dominant class ("Baseline"). This ensures that the model learns to identify the specific physiological signatures of "Amusement" and "Meditation", resulting in more equitable decision boundaries and a higher Macro F1-score.

3.4 Improvements

To combat the inherent class imbalance (where "Amusement" has fewer samples), we apply **SMOTE (Synthetic Minority Over-sampling Technique)** to the training set, creating synthetic examples of the minority classes to balance the decision boundaries.

For the Random Forest Classifier, we configured the model to use a **random subset of features (square root of the total count)** at each split to prevent any single dominant feature from overshadowing other subtle but important HRV markers.

For the deep learning models, we inject **Gaussian noise** and **random amplitude scaling (0.9x to 1.1x)** during training to simulate sensor variability. This data augmentation strengthens the models' resilience by forcing the networks to learn robust morphological patterns rather than memorizing specific signal intensities or noise profiles from the training set.

4 Experiments and Results

In section 3, we trained 6 different models—SVM, Random Forest, 1D CNN, CNN-LSTM, LOSO SVM, and LOSO RF. In this section, we present the quantitative evaluation of our models using subject-independent testing protocols and analyze the resulting performance metrics and confusion matrices.

4.1 Evaluation Metrics

To assess the performance of our models, we utilize two primary metrics: **Accuracy** and the **Macro-averaged F1-Score**.

Accuracy. This measures the total percentage of correctly classified windows across all four emotional states. While useful, accuracy can be misleading in the WESAD dataset because the classes are imbalanced. For instance, "baseline" data is significantly more abundant than "Amusement".

Macro F1-Score. This is our most critical metric. The Macro F1-score calculates the F1-score for each emotional class independently and then averages them. This ensures that the model's ability to identify the minority class ("Amusement") is weighted equally with the majority class ("Baseline"). A high Macro F1-score indicates that the model is a robust emotion recognizer, not just a "majority class predictor".

4.2 Main Results

The performance of our six models is summarized in the table below. These results represent testing on subjects the models have never seen before (S2, S3, and S4 for the fixed split, and every subject individually for LOSO).

Model	Testing Accuracy	Macro F1-Score
SVM (RBF Kernel)	0.5418	0.5053
Random Forest	0.6364	0.5564
1D CNN	0.6545	0.5982
CNN-LSTM	0.6400	0.5321
LOSO SVM	0.5722	0.5127
LOSO RF	0.6028	0.4855

Table 1. Model Performance Comparison Table

The **1D CNN** emerged as the top-performing model, achieving the highest accuracy (65.45%) and macro f1-score (0.5982). This suggests that learning features directly from the filtered BVP waveform (end-to-end) is more effective than relying on handcrafted HRV features (feature-based). Among the feature-based models, the **Random Forest** outperformed the SVM, likely due to its ability to handle the non-linear thresholds associated with physiological shifts. The **LOSO (Leave-One-Subject-Out)** results show a slight performance decrease compared to the fixed, which is an expected "generalization gap" when the model must adapt to the unique vascular characteristics of a brand-new user.

4.3 Training Dynamics

The loss curves (Figures 2-4) for the 1D CNN, LSTM, and CNN-LSTM models demonstrate successful convergence over 60 epochs.

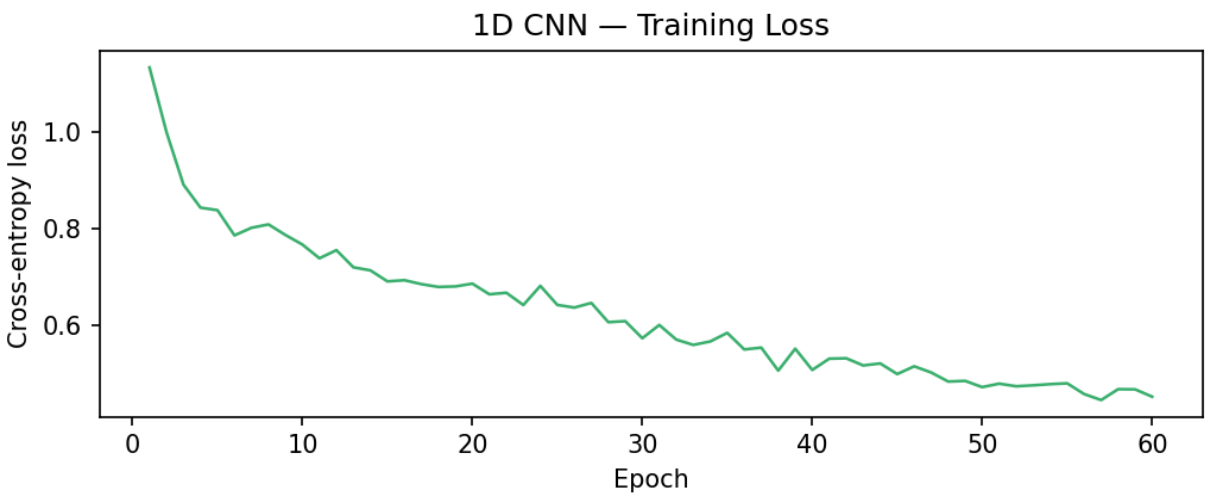
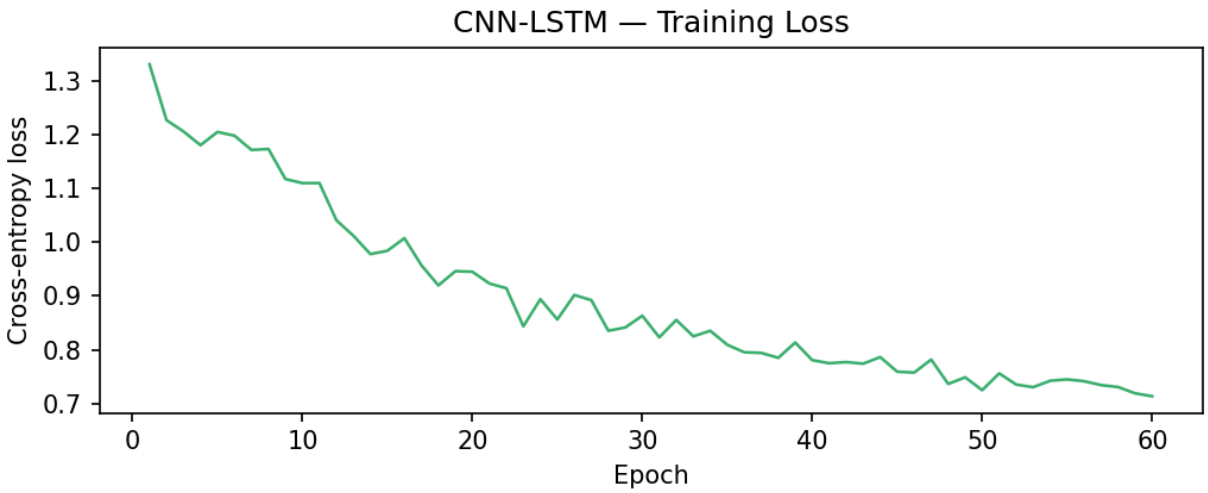
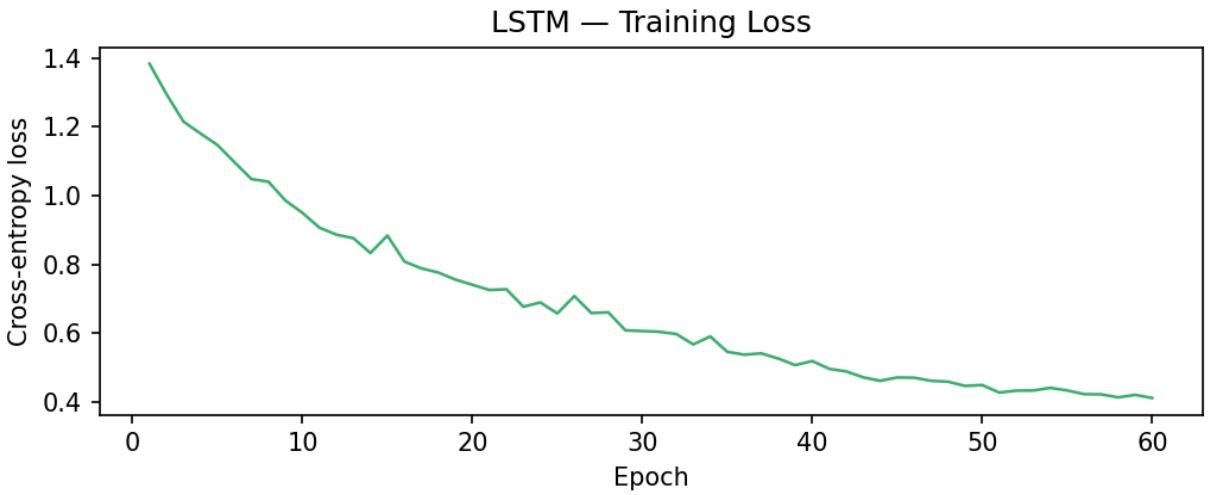


Figure 2. The **1D CNN** exhibited a stable decay, suggesting that its spatial filters quickly identified robust features in the BVP signal.





Figures 3&4. The **CNN-LSTM** and **LSTM** curves showed higher volatility (especially CNN-LSTM), reflecting the inherent difficulty in stabilizing gradients within recurrent architectures when processing long physiological sequences.

The contrast between the terminal losses of our three deep learning architectures reveals a significant trade-off between model capacity and ease of optimization. Interestingly, the **standalone LSTM** achieved the lowest overall terminal loss (~0.40), suggesting that its recurrent structure is highly efficient at minimizing cross-entropy error on the training set. However, when comparing this to our final results table, we observe that this low training loss did not translate into the highest testing F1-score. This suggests that the LSTM may have been "over-optimizing" for the specific temporal sequences in the training data, a common challenge in physiological signal processing where the model can easily overfit to subject-specific heart rate rhythms.

In contrast, the **CNN-LSTM hybrid** maintained a much higher terminal loss (~0.71). This higher loss reflects the increased complexity of the hybrid architecture, which must simultaneously optimize the spatial filters of the CNN encoders and the temporal memory of the LSTM layers. The "jagged" nature of the hybrid's loss curve indicates that the optimizer faced a more difficult loss landscape, as it struggled to align the features extracted from one-second chunks with the long-term emotional context of the 60-second window.

The **1D CNN** provided the most balanced training profile, with a terminal loss of approximately 0.45 and the most stable, consistent decay. This stability is likely why the 1D CNN achieved the highest generalization (65.45% accuracy). While the LSTM was more aggressive at minimizing loss, the 1D CNN's spatial approach—facilitated by **Batch Normalization** and **Global Average Pooling**—provided a "smoother" learning process.

The effectiveness of the **Cosine Annealing** and **Gaussian noise injection** (mentioned in the Improvements section) acted as effective regularizers, as it is clearly visible in the latter half of the training processes (epochs 40-60), where oscillations begin to dampen. As the learning rate decayed toward its minimum, the models were able to perform finer weight updates, allowing them to settle into a tighter local minimum. This transition from high exploration to high exploitation was essential for the LSTM hidden states to stabilize, ensuring that the final model was not over-sensitized to the most recent batches of training data, but had truly learned a generalized representation of the emotional classes.

4.4 Confusion Matrix Analysis

A deep dive into the confusion matrices (Figures 5-7) of our top-performing model, our most generalized model, and our hybrid model helps us understand the physiological intersections between emotional classes.

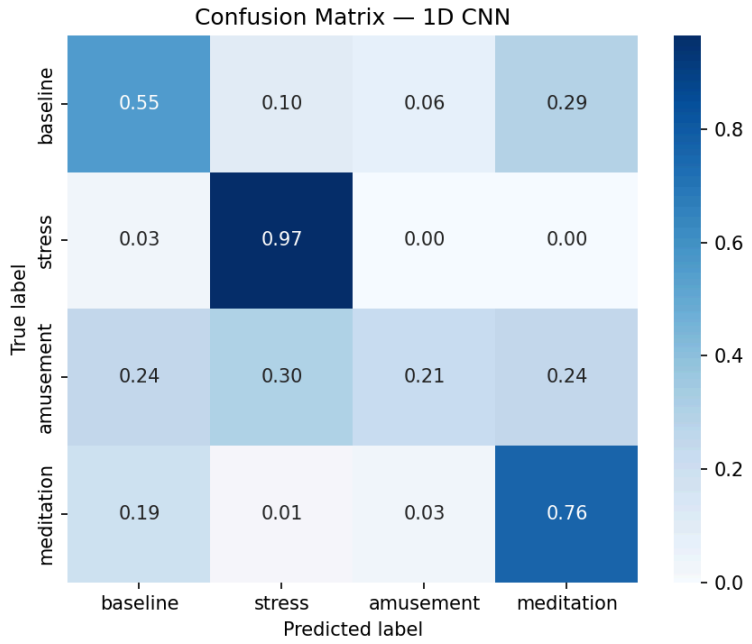


Figure 5. Confusion matrix for **1D CNN**

The High-Arousal Success: 1D CNN. The confusion matrix for the **1D CNN** (Figure 5) highlights the model's exceptional ability to isolate high-arousal negative states. The model achieved a **97% recall for Stress**, misclassifying only 3% of stress windows as baseline. This suggests that the morphological changes in the PPG wave during stress—likely characterized by increased vascular resistance and a simplified pulse shape—are highly distinct and easily captured by convolutional filters. However, the model struggled with Amusement (21% recall), with 30% of amusement windows being misclassified as Stress. This "Arousal Confusion" is a known phenomenon where positive and negative high-arousal states exhibit similar heart rate elevations, making them difficult to distinguish using PPG alone.

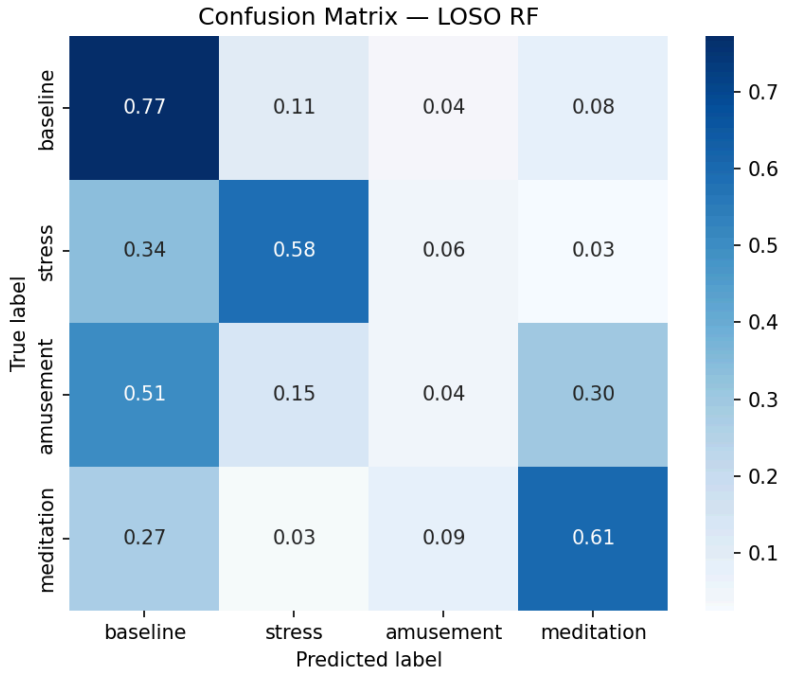


Figure 6. Confusion matrix for **LOSO RF**

The Generalization Gap: LOSO Random Forest. The **LOSO RF** matrix (Figure 6) represents the model's performance on a completely unseen subject. Here, we see a significant "smoothing" of the diagonal. The accuracy for stress drops from the 1D CNN's 97% **down to 58%**. Interestingly, the model defaults heavily to **Baseline (77% accuracy)**. This indicates that when the model encounters a brand-new physiological profile, it tends to be conservative, labeling ambiguous signals as "Neutral/Baseline." The near-total failure to identify **Amusement (4%)** in the LOSO setup proves that the markers for positive emotions are highly subject-specific and do not generalize well across a broad population without personalized calibration.

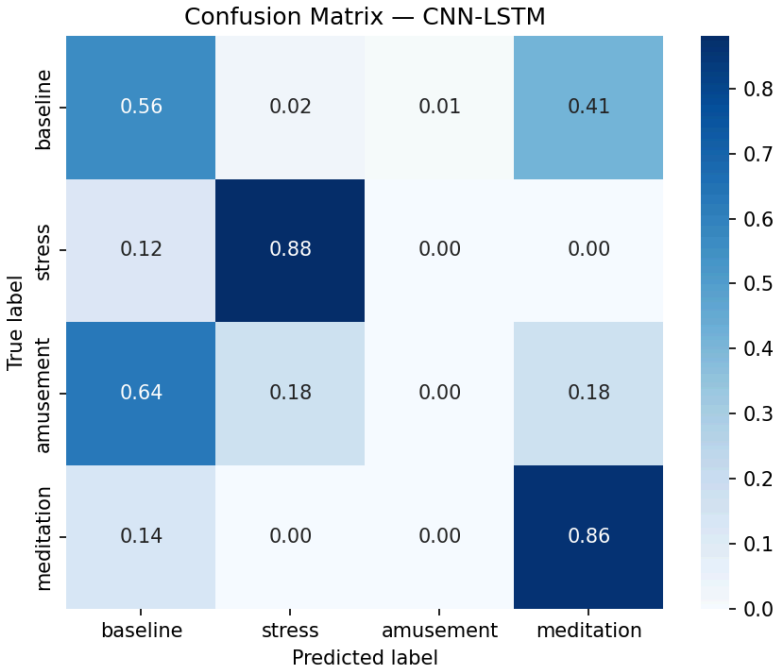


Figure 7. Confusion matrix for **CNN-LSTM**

The Sequence Failure: CNN-LSTM. The **CNN-LSTM** matrix (Figure 7) provides a fascinating look at the failure of temporal modeling for specific classes. While the model performed excellently on **Meditation (86% recall)** (even outperforming the 1D CNN), it **completely failed to identify Amusement (0.00%)**. Every single amusement window was misclassified as either Baseline (64%) or Meditation (18%). This suggests that while the LSTM is excellent at identifying the long-term, slow rhythmic oscillations associated with deep relaxation (Meditation), it "smooths over" the short-lived, erratic pulse variations that characterize Amusement, effectively treating them as baseline noise.

The disparity in these matrices shows that different models are experts in different emotions. The 1D CNN is the expert in Stress, the CNN-LSTM in Meditation, while the RF is in the Baseline state. This suggests that a future ensemble approach that combines the strengths of these architectures can be the key to achieving universal accuracy in wearable emotion recognition.

4.5 Feature Importance Analysis

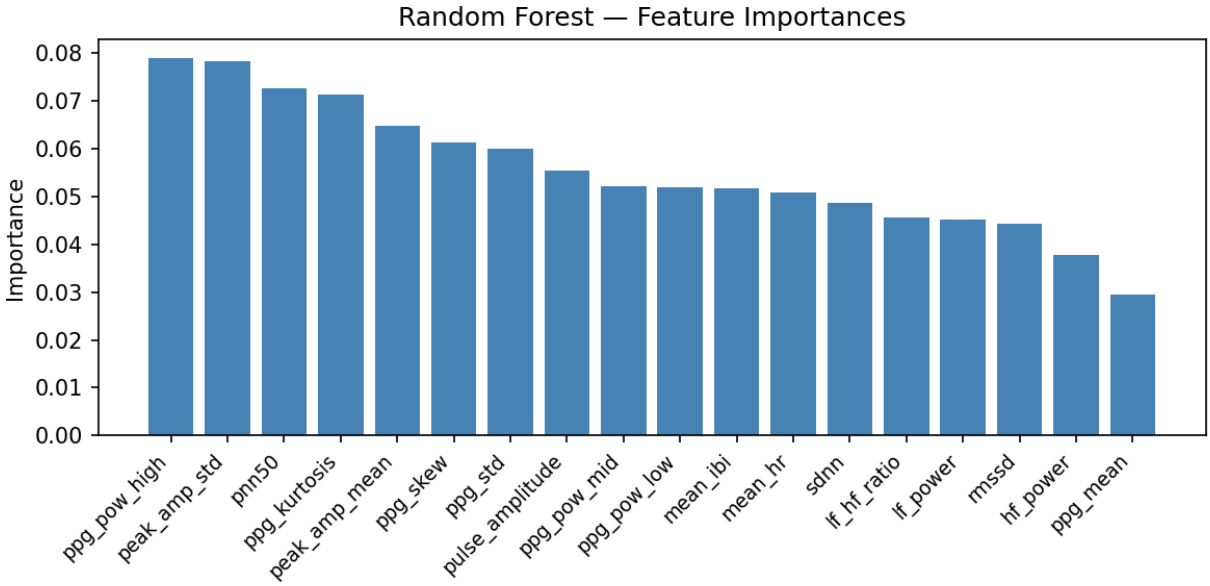


Figure 8. Feature Importance Plot for **Random Forest**

The **Random Forest** feature importance plot (Figure 8) provides a window into the physiological markers of emotion through a feature-based approach. The most important feature was **ppg_pow_high** (spectral power in the 2.0-4.0 Hz range), followed by **peak_amp_std** (variability in pulse height). This indicates that the morphology and spectral distribution are more predictive of emotion than traditional metrics like **mean_ibi** or **mean_hr**, which ranked in the middle of the plot. This validates our decision to adopt an additional end-to-end deep learning approach, as the CNN is naturally designed to extract these high-frequency morphological features that standard HRV features might overlook.

5 Discussion

In this section, we interpret our findings, weighing the tradeoffs between model complexity and interpretability, and acknowledging the constraints of the current study.

5.1 Key Findings

Our experiments yielded several high-level insights that challenge traditional assumptions in physiological signal processing:

Morphology Trumps Statistics in Stress Detection: The most striking finding was the **97% recall rate for Stress achieved by the 1D CNN**, compared to the much lower performance of the feature-based Random Forest. This suggests that the physiological "signature" of stress is not found in the HRV features, but in the micro-deformations of the pulse wave shape itself. The CNN was able to detect subtle changes in vascular resistance that summary statistics like Mean HR simply cannot quantify.

Vascular Tone is a Primary Emotional Marker: Our feature importance analysis revealed that **ppg_pow_high (2.0–4.0 Hz spectral power)** was the most predictive feature, outperforming Heart Rate. This indicates that the "vibration" or spectral richness of the blood volume pulse is a more sensitive marker for emotional shifts than the pulse rate. This finding suggests that the autonomic nervous system's impact on peripheral vascular tone (vasoconstriction) is a more reliable emotional indicator than its impact on the sinoatrial node (heart rate rhythm) when using wrist-worn sensors.

The Temporal Modeling Duality: We discovered a clear split in the effectiveness of the **LSTM** architecture. While the LSTM was our best model for identifying **Meditation (86% recall)**, it was our worst for **Amusement (0% recall)**. This reveals a fundamental characteristic of the models: LSTMs are "experts" at identifying slow-moving, rhythmic physiological transitions (the long, deep breathing cycles of meditation) but act as "low-pass filters" that smooth over the erratic, short-lived "bursts" of cardiac activity that define amusement.

The Subject-Independence Barrier: The significant performance gap between our **fixed-subject split (65.4% accuracy)** and our **LOSO-CV results (60.2% for RF)** highlights that **physiological "identity" is often stronger than emotional "signal"**. Our models frequently had to fight against the unique "cardiac fingerprint" of each subject to find the underlying emotional state, confirming that universal emotion recognition remains a challenge of filtering out individual identity to find collective humanity.

5.2 Trade-offs

Our results highlight a clear trade-off between model interpretability and raw performance. **Feature-Based Models (SVM/RF):** These models provide clinical transparency. Using the Random Forest, we can pinpoint exactly which HRV markers are shifting, which is essential for medical applications where a "why" is required for a diagnosis. However, these models are limited by the quality of peak detection; in noisy segments where peaks cannot be found, these models lose data.

Deep Learning Models (CNN/LSTM): These models are far more resilient to noise and achieved the highest F1-scores. They do not require manual peak detection, making them better

suited for the "messy" data typical of ubiquitous sensing. However, they act as "black boxes," making it difficult to understand the underlying physiological rationale for a classification.

Temporal Modeling: The CNN-LSTM hybrid showed that while LSTMs are excellent at identifying slow-moving states like Meditation (86% recall), they tend to "smooth over" shorter emotional bursts, leading to the total failure in Amusement detection.

5.3 Limitations

The primary limitation of this study is the **inter-subject variability**, as evidenced by the performance drop in our LOSO experiments. Physiological responses are highly individual, where a heart rate that signifies "Stress" for one person may be "Baseline" for another. Additionally, we focused exclusively on the PPG (BVP) signal. While PPG is convenient for wrist-worn devices, it is highly sensitive to motion artifacts. In a real-world setting (outside the controlled WESAD laboratory environment), physical movement would introduce significant noise that a 0.5–4.0 Hz filter might not fully remove.
Finally, the small sample size of 15 subjects limits the depth to which a deep neural network can generalize across the entire human population.

5.4 Future Work

The exceptional performance of our **1D CNN in Stress detection (97% recall)** opens a potential pathway for specialized health applications. Future research and development could involve the following areas:

Multimodal Sensor Fusion: While PPG provides a rich morphological signature, it inherently struggles with “Arousal Confusion”. To solve this, future models should explore **Multimodal Fusion**, integrating **Electrodermal Activity** and **Accelerometer data**. EDA is a pure measure of sympathetic activation and could act as a verification to distinguish between amusement-induced arousal and genuine distress. Furthermore, using accelerometer data as a noise-reference signal would allow for **Adaptive Filtering**, which could subtract motion-induced noise from the PPG signal, significantly increasing the model’s accuracy.

Specialized Stress-Monitoring and Intervention Tools: Our model’s high sensitivity to stress makes it an ideal candidate for warning systems in high-stakes occupational environments, such as for healthcare workers or first responders. Future work could focus on optimizing the 1D CNN for **real-time** and **on-device processing** within mobile health platforms. This enables the system to not only detect stress and burnout in early stages but also trigger interventions (such as prompting the user to engage in biofeedback exercises to downregulate the sympathetic nervous system before the stress becomes chronic).

Long-Term Temporal Trend Analysis: While our current windows are 60 seconds long, future work could examine **circadian and long-term trends**. By aggregating stress detections over hours or days, we could develop models that distinguish between **"Acute Stress"** (temporary challenges) and **"Chronic Stress"** (long-term physiological strain). This would involve adding a secondary, higher-level temporal layer to analyze the frequency and persistence of stress events throughout a whole week.

6 Conclusion

In this final section, we summarize the project's contributions and reflect on the viability of our modeling architectures for the task of automated emotion recognition.

This project provided a rigorous comparative analysis of four distinct modeling paradigms: RBF-kernel SVMs, Random Forests, 1D CNNs, and CNN-LSTM hybrids. Our results demonstrate that while traditional machine learning models (SVM and Random Forest) offer a strong baseline and high interpretability, they are ultimately limited by the constraints of handcrafted feature engineering. The Random Forest achieved a respectable 63.6% accuracy, but it was unable to capture the nuanced morphological signatures that define high-arousal states, as seen by its significantly lower recall for stress compared to deep learning alternatives.

The transition to an **end-to-end 1D CNN** model proved to be the most significant architectural advancement in our pipeline. By bypassing manual feature extraction and learning filters directly from the filtered waveform, the 1D CNN achieved a **peak accuracy of 65.45% and a nearly perfect 97% recall for stress**. This confirms that convolutional architectures are uniquely suited for physiological signals, where the spatial arrangement of a pulse wave—its peaks, troughs, and slopes—contains more discriminative power than summary statistics. Furthermore, our experiments with **CNN-LSTM hybrids** revealed that while temporal modeling is highly effective for identifying stable, long-term states like meditation (86% recall), it can lead to a "smoothing effect" that renders the model blind to shorter, more erratic emotional bursts like amusement.

Ultimately, this study concludes that the **1D CNN is the most viable model** for subject-independent emotion recognition due to its superior balance of accuracy and generalization. Our use of **LOSO cross-validation** served as a necessary "reality check," highlighting that while deep learning can achieve high performance on fixed splits, the true challenge lies in architectural generalizability across diverse physiological profiles. This work establishes a clear modeling hierarchy, suggesting that future affective computing systems should prioritize convolutional feature learning to navigate the complex, personal, and often noisy landscape of human emotional data.

Omitted Results

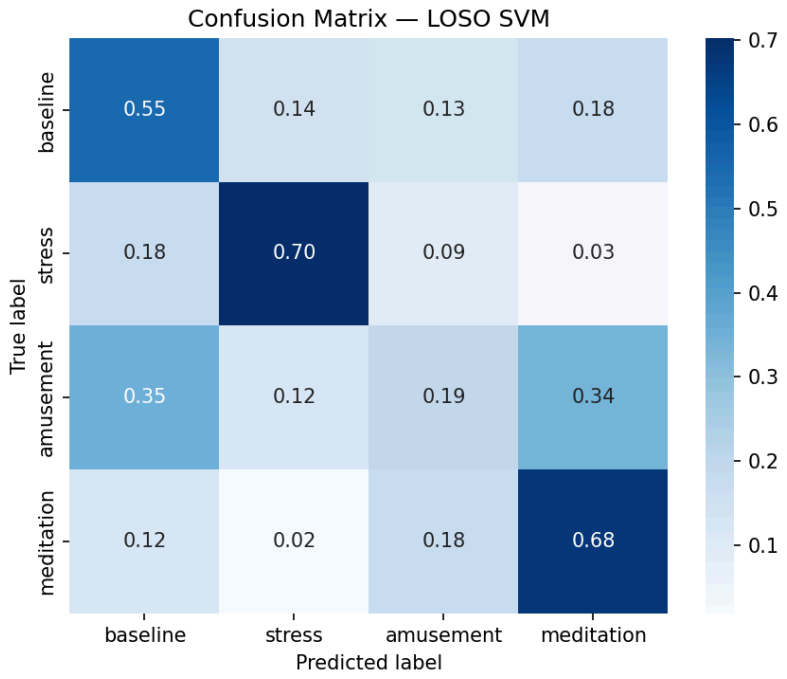


Figure 9. Confusion matrix for **LOSO SVM**

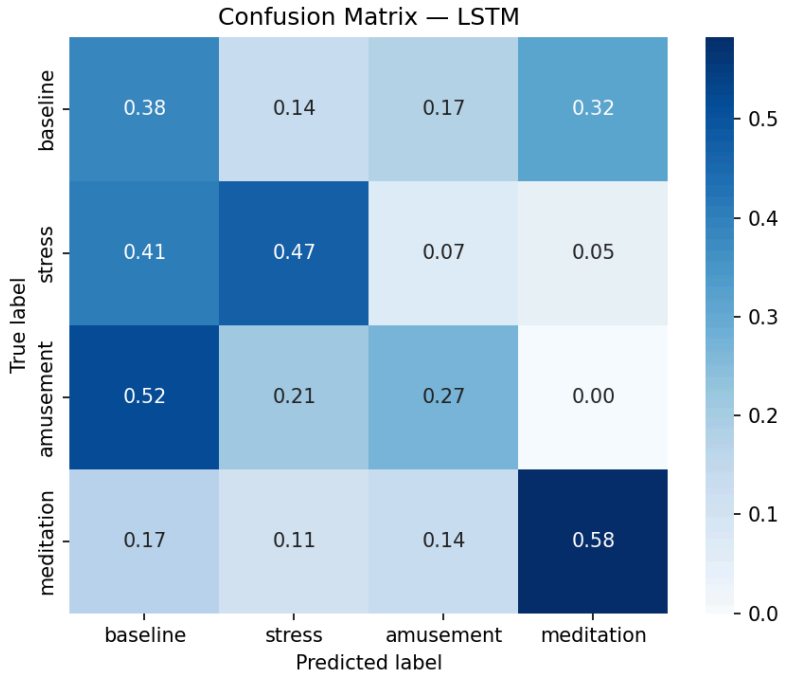


Figure 10. Confusion matrix for **LSTM**

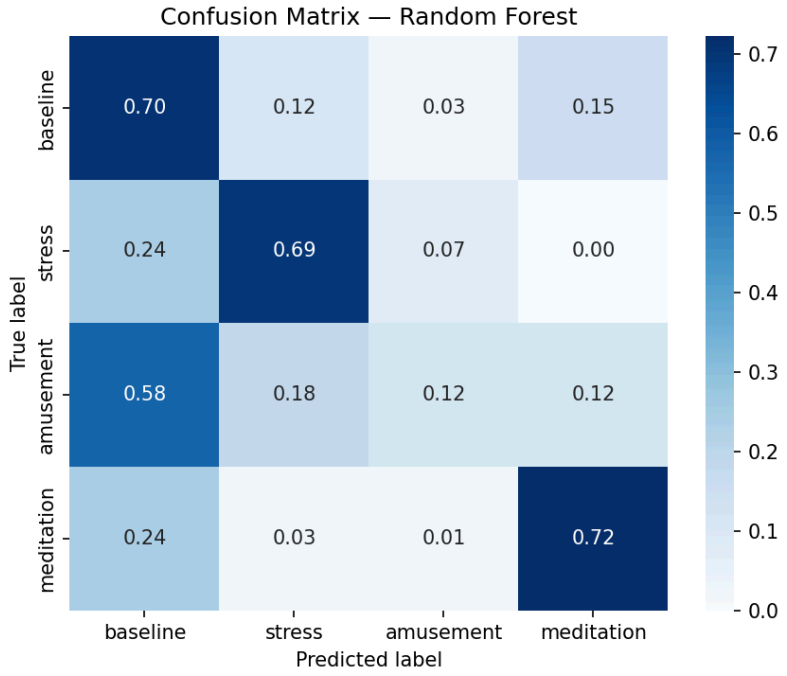


Figure 11. Confusion matrix for **RF**

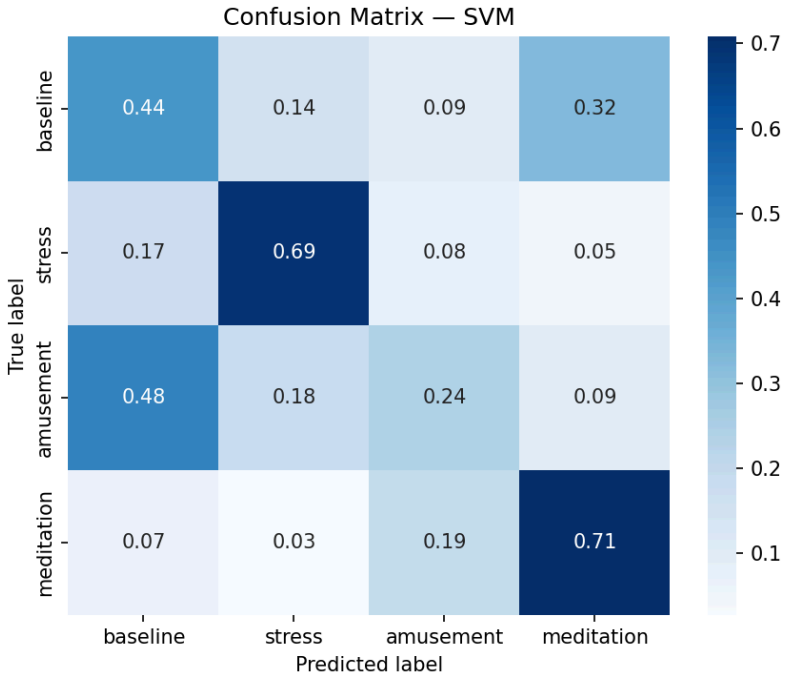


Figure 12. Confusion matrix for **SVM**